Data types with



Navigate to the **03-Types** folder.

Open **03-Types-Exercises.Rmd**

Recall

What types of data are in this data set?

	time_hour	name	air_time	distance	day	delayed
1	2013-01-01 05:00:00	United Air Lines Inc.	13620s (~3.78 hours)	1400	Tuesday	TRUE
2	2013-01-01 05:00:00	United Air Lines Inc.	13620s (~3.78 hours)	1416	Tuesday	TRUE
3	2013-01-01 05:00:00	American Airlines Inc.	9600s (~2.67 hours)	1089	Tuesday	TRUE
4	2013-01-01 05:00:00	JetBlue Airways	10980s (~3.05 hours)	1576	Tuesday	FALSE
5	2013-01-01 06:00:00	Delta Air Lines Inc.	6960s (~1.93 hours)	762	Tuesday	FALSE
6	2013-01-01 05:00:00	United Air Lines Inc.	9000s (~2.5 hours)	719	Tuesday	TRUE
7	2013-01-01 06:00:00	JetBlue Airways	9480s (~2.63 hours)	1065	Tuesday	TRUE
8	2013-01-01 06:00:00	ExpressJet Airlines Inc.	3180s (~53 minutes)	229	Tuesday	FALSE
9	2013-01-01 06:00:00	JetBlue Airways	8400s (~2.33 hours)	944	Tuesday	FALSE
10	2013-01-01 06:00:00	American Airlines Inc.	8280s (~2.3 hours)	733	Tuesday	TRUE
11	2013-01-01 06:00:00	JetBlue Airways	8940s (~2.48 hours)	1028	Tuesday	FALSE

Common data types

1. Logicals
2. Strings
3. Factors
4. Dates and Times



Logicals



Logicals

R's data type for boolean values (i.e. **TRUE** and **FALSE**).

```
TRUE
##  TRUE

typeof(TRUE)
##  "logical"

typeof(c(TRUE, TRUE, FALSE))
##  "logical"
```



Details

⚠ The class of service you searched may not be available on one or more flights

BNA - ORD
Flight 1 of 2

ORD - YVR
Flight 2 of 2

Nashville, TN to Chicago, IL

Thursday, July 26, 2018

4:10 PM → 6:03 PM

AA 3246 ▪ CR7-Canadair RJ 700

Operated by SkyWest Airlines As American Eagle

Travel info

Travel time: **1h 53m**
Connection time: **2h 33m**

Performanc

On time: **52**
Late: **43%**

Main Cabin

Meals: Beverage service
Booking code: V
Class: Economy

Business

Meals: Beverage service
Booking code: I
Class: First

* This is based on information from the month of May 2018
** The on-time arrival percentage for the selected flight is based on arrival within 14 minutes after

Performance*

On time: **52%****
Late: **43%**

** The on-time arrival percentage for the selected flight is based on arrival within 14 minutes after the scheduled arrival as reported monthly to the U.S. Department of Transportation.

Warm Up

Did you fly here?

Did your flight arrive late?

```
flights %>%
  mutate(delayed = arr_delay > 0) %>%
  select(arr_delay, delayed)
```

arr_delay <dbl>	delayed <lgl>
11	TRUE
20	TRUE
33	TRUE
-18	FALSE
-25	FALSE
12	TRUE
19	TRUE
-14	FALSE
-8	FALSE
8	TRUE



```
flights %>%
  mutate(delayed = arr_delay > 0) %>%
  select(arr_delay, delayed)
```

arr_delay <dbl>	delayed <lgl>
11	TRUE
20	TRUE
33	TRUE
-18	FALSE
-25	FALSE
12	TRUE
19	TRUE
-14	FALSE
-8	FALSE
8	TRUE

Can we compute the proportion of NYC flights that arrived late?



Most useful skills

1. Logical tests
2. Math with logicals



Quiz

What does this return?

TRUE + 0

NOT AN ERROR!

Quiz

What does this return?

TRUE + 0

1

Math with logicals

TRUE = 1

FALSE = 0



Quiz

What will this return?

sum(c(FALSE, FALSE, TRUE, FALSE)**)**

THE NUMBER OF TRUES

Quiz

What will this tell us?

sum(flights$arr_delay > 0**)**

THE NUMBER OF DELAYED FLIGHTS

Quiz

What will this tell us?

mean(flights$arr_delay > 0**)**

THE PROPORTION OF DELAYED FLIGHTS

TRUE = 1

FALSE = 0

sum() = **number** that pass

mean() = **proportion** that pass



Your Turn 1

Use flights to create **delayed**, a variable that displays whether a flight was delayed (arr_delay > 0).

Then, filter to rows where **delayed** does not equal NA.

Finally, create a summary table that shows:

1. How many flights were delayed
2. What proportion of flights were delayed



```
flights %>%
  mutate(delayed = arr_delay > 0) %>%
  filter(!is.na(delayed)) %>%
  summarise(total = sum(delayed), prop = mean(delayed))
## # A tibble: 1 × 2
##   total      prop
##   <int>     <dbl>
## 1 133004 0.4063101
```



Strings



(character) **strings**

Anything surrounded by quotes(") or single quotes(').

```
"one"
##  "one"

typeof("one")
##  "character"

typeof("oops. I'm stuck in a string)
+ ")
##  "character"
```




Warm Up

Decide in your group:

Are boys names or girls names more likely to end in a vowel?



babynames



Names of male and female babies born in the US from 1880 to 2017. 1.9M rows.

```
# install.packages("babynames")
library(babynames)
```

babynames

year <dbl>	sex <chr>	name <chr>	n <int>	prop <dbl>
1880	F	Mary	7065	7.238433e-02
1880	F	Anna	2604	2.667923e-02
1880	F	Emma		
1880	F	Elizabeth		
1880	F	Minnie		
1880	F	Margaret		
1880	F	Ida		
1880	F	Alice	1414	1.448711e-02
1880	F	Bertha	1320	1.352404e-02
1880	F	Sarah	1288	1.319618e-02

1–10 of 1,858,689 rows Previous 1 2 3 4 5 6 ... 100 Next

How can we build the proportion of boys and girls whose name ends in a vowel?



Most useful skills

1. How to extract/ replace substrings
2. How to find matches for patterns
3. Regular expressions



stringr



Simple, consistent functions for working with strings.

```
# install.packages("tidyverse")
library(tidyverse)
```




str_sub()

Extract or replace portions of a string with **str_sub()**

```
str_sub(string, start = 1, end = -1)
```

string(s) to manipulate

position of first character to extract within each string

position of last character to extract within each string



Quiz

What will this return?

```
str_sub("Garrett", 1, 2)
```

Quiz

What will this return?

```
str_sub("Garrett", 1, 2)
```

"Ga"

Quiz

What will this return?

```
str_sub("Garrett", 1, 1)
```

Quiz

What will this return?

```
str_sub("Garrett", 1, 1)
```

"G"

Quiz

What will this return?

```
str_sub("Garrett", 2)
```

Quiz

What will this return?

```
str_sub("Garrett", 2)
```

```
"arrett"
```

Quiz

What will this return?

```
str_sub("Garrett", -3)
```

Quiz

What will this return?

```
str_sub("Garrett", -3)
```

```
"ett"
```

Quiz

What will this return?

```
g <- "Garrett"
str_sub(g, -3) <- "eth"
g
```

Quiz

What will this return?

```
g <- "Garrett"
str_sub(g, -3) <- "eth"
g
```

```
"Garreth"
```

Your Turn 2

In your group, fill in the blanks to:

1. Isolate the last letter of every name
2. and create a logical variable that displays whether the last letter is one of "a", "e", "i", "o", "u", or "y".
3. Use a weighted mean to calculate the proportion of children whose name ends in a vowel (by year and sex)
4. and then display the results as a line plot.

